

CSCIM_FS: Cosine similarity coefficient and information measurement criterion-based feature selection method for high-dimensional data



Gaoteng Yuan^{a,*}, Yi Zhai^b, Jiansong Tang^a, Xiaofeng Zhou^a

^a College of Computer and Information, Hohai University, Nanjing 211100, China

^b School of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), Jinan 250353, China

ARTICLE INFO

Article history:

Received 29 March 2023

Revised 15 June 2023

Accepted 8 July 2023

Available online 13 July 2023

Keywords:

Feature selection
Mutual information
Cosine transform
Classification

ABSTRACT

Feature selection (FS) based on mutual information (MI) metrics needs to discretize the data in pre-processing, which is a convenient way to identify correlation between features. However, information loss often occurs in data discretization. In order to solve this information loss problem, this paper proposes a FS algorithm based on cosine similarity coefficient and information measurement criterion (CSCIM_FS). First, the MI between features and tags is calculated, and features are sorted out according to the MI calculated. Then, a feature matrix is constructed to transform the one-dimensional feature sequence into a two-dimensional square matrix. Next, cosine transform is adopted to obtain the high-frequency components of the feature matrix, and sampling is conducted to derive the hash fingerprint of the feature matrix. After that, the similarity between every two features is calculated on the basis of the hash fingerprints of different features. Finally, the feature weight is calculated according to tags, the MI and similarity between features, and a key feature subset is obtained and used to conduct feature selection from the data. The experimental results on several UCI public datasets show that CSCIM_FS algorithm selected a feature subset with high accuracy, and that this algorithm performs better than MIM, CMIM, mRMR and other algorithms.

© 2023 Elsevier B.V. All rights reserved.

1. Introduction

The main purpose of feature selection (FS) is to select key features from the original feature set and to exclude unnecessary features so as to improve the efficiency and accuracy of the classification model [1]. FS algorithm needs to balance the contradiction between feature subset size and task accuracy as much as possible. Meanwhile, it's expected to exhibit timeliness and reduce time complexity and space complexity [2]. As one of the most heated research topics in machine learning, data mining and pattern recognition, FS algorithms have been widely studied in past few decades [3,4].

From the perspective of algorithm design, according to the key feature subset searching, FS algorithms are roughly divided into three broad categories, namely filters, wrapper models and embedded techniques [5–7]. Different FS methods work for different occasions. The filter-based FS process is independent of the subsequent classification model, which filters feature set according to a class label measurement criteria, and then train classification model [8–10]. The common measurement criteria of filter-based

FS include distance, chi square statistics, information theory and so on [11,12]. Unlike the filter-based FS, FS using wrapper method takes the classification model to be used as the evaluation criterion of feature subset. That is, the classifier is provided in advance, then optimization is performed according to the specified classifier. Commonly used searching strategies of wrapper methods include genetic algorithm, simulated annealing algorithm and so on [13,14]. However, in the process of FS using wrapper methods, the classifier needs to be trained by the algorithm for several times, so the computational cost is very high [7]. Embedded methods can be regarded as a compromise between filter-based FS and FS using wrapper methods. The basic idea of embedded methods is to embed the FS process in the algorithm. When the training process of classification algorithm is completed, the feature subset is obtained [15]. The commonly used embedded methods for FS include CART and C4.5 algorithm, etc [16,17]. Compared with embedded techniques, wrapper models are more direct and targeted, but they have higher algorithm complexity. In contrast, filter-based FS is independent of the predetermined learning algorithm and has stronger generalization. And due to its low computational complexity, it is suitable for processing high-dimensional data. Since this paper mainly focuses on FS for high-dimensional data processing, so we will mainly talk about filter-based FS.

* Corresponding author.

Many filter-based FS methods are developed according to information theory which is widely used in FS [18]. Mutual information (MI)-based filter methods have gained popularity due to their ability in capturing the non-linear association between dependent and independent variables in machine learning. MI strategy is employed to analyze the correlation between features and category labels, thereby completing the selecting of key feature subsets [19,20]. However, before constructing the FS method based on mutual information measurement criteria, it is necessary to discretize the features and count the distribution probability of features [21], where a lot of information is lost. Moreover, when judging the redundancy between features, it's important to compare the feature sequences consisting of different feature variables. Since one-dimensional feature sequences make it difficult to conduct multiple-angle analysis when judging whether the features are redundant, which then affects the performance of FS algorithm [22].

To address the above mentioned problems, we propose a FS algorithm for high-dimensional data based on cosine similarity coefficient and information measurement criterion (CSCIM_FS), which uses MI strategy to select the features that are highly correlated with classification labels. In addition, a two-dimensional matrix is developed for original feature sequence, and the redundancy between features is eliminated according to the cosine similarity between features, so as to construct the feature evaluation function and complete the screening of key feature subsets. The experiments on several UCL datasets show that CSCIM_FS algorithm can make up for the lost information during data discretization, and that the accuracy of redundant features identification is better than most baselines which depends only on MI strategy. These results indicate that our algorithm exhibits very strong competitiveness.

The remainder of this paper is organized as follows: Section 2 discusses related works. Section 3 introduces basic theories involved in the algorithm. The construction of CSCIM_FS algorithm is shown in Section 4. Section 5 describes the data set, the evaluation indices, experimental process and experimental results. Conclusions and future research directions are presented in Section 6.

2. Related work

Two contents are introduced in this section. One is the FS algorithm based on MI strategy and the other is the similarity measurement method. These two concepts are introduced to facilitate the subsequent analysis of the advantages and disadvantages of CSCIM_FS algorithm.

2.1. Summary of FS methods based on MI strategy

FS method mainly achieve the following two purposes: maximizing the correlation between feature subsets and target categories while minimizing the redundancy between feature subsets [1,23]. The search of the key feature subset can be divided into two steps. First, various metrics are used to define the dependencies between features and classes, such as distance, consistency, dependencies and information. Then, the evaluation criteria based on the above measurement are set and used to evaluate the advantages and disadvantages of a single feature or a feature subset. Through the continuous iteration of the above two steps, optimal feature subset screening is finally achieved. MI is a popular information measurement criterion, which is widely adopted in FS algorithm. The earliest FS algorithm based on MI strategy is MIM (mutual information maximum) algorithm [24], which calculates the MI between features and tags in turn, then sorts out these features to obtain the key feature subset. However, the algorithm is

unable to identify the redundant features in the data, and overfitting usually occurs in model training, so the algorithm suffers from limitations. In order to improve the shortcomings of MIM algorithm and eliminate redundant features during feature selection, Battiti et al. [25]. Proposed MIFS (MI-haled feature selection) algorithm. By introducing a first-order incremental search algorithm and using greedy selection method to filter feature subsets, the redundant features in MIM algorithm can be removed. However, when the algorithm measures the redundancy between features, it is likely to introduce many irrelevant features, so it is necessary to find ways to remove these unnecessary features. To this end, Peng et al. [26]. improved the MIFS algorithm and proposed the mRMR (maximum relevance minimum redundancy) algorithm. The algorithm updated the evaluation criteria, taking the MI between the candidate features and the class label as the correlation measurement criterion, and the MI between the candidate features and the selected features as the redundancy measurement criterion. Moreover, by weighing the maximum difference between the two criteria in the key feature subset screening, the algorithm can achieve good results. However, mRMR algorithm only considers the relationship between two variables and tags, and it does not examine multiple variables. In order to clarify the relationship between multiple variables, the JMI (joint mutual information) algorithm was proposed by Yang et al. [27]. In this algorithm, the conditional MI strategy is used to calculate the relationship between multiple variables and data labels by calculating the conditional MI after determining two variables, so as to improve the accuracy of feature selection. A similar algorithm is CIFE (conditional information feature extraction) algorithm developed by Lin et al. [28]. In addition, many FS models combine MI strategy with intelligent algorithm, such as MI and swarm intelligence algorithm, particle swarm optimization with MI, genetic algorithm with MI, and so on [29], which achieved good results in corresponding applications. Although the above algorithms performed well in their corresponding data sets, the similarity between features cannot be fully evaluated solely by a single MI strategy. In particular, feature discretization before constructing the FS model using MI strategy loses a lot of valuable information, which has a great impact on the FS algorithm, so we need to solve this problem.

MI strategy-based FS algorithm primarily solves two problems, including how to find out relevant features and how to remove redundant features. Because of the differences between datasets and the application fields of different algorithms, the improvement of MI strategy-based FS algorithm has been a research hotspot.

2.2. Summary of similarity measurement methods

Similarity measurement is a hot research topic in recommendation system development. It is mainly used for image and text analysis. In image registration and semantic matching, similarity measurement permits well summarization and analysis of the data. Commonly used similarity calculation methods include distance, Pearson correlation coefficient, mutual information, and cosine similarity, etc [30–32]. For example, Yu et al. [33]. presented three quantum algorithms based on Euclidean distance to measure the similarity between datasets, which were verified on several datasets and showed good measurement results. However, the distance based measurement criterion cannot be used to calculate the similarity of multimodal data. In order to explore the power consumption patterns, Lin et al. [34]. proposed a weighted Pearson hierarchical clustering algorithm based on the weighted Pearson distance which is used to calculate the similarity between datasets, and this algorithm has achieved higher clustering accuracy. Ref. [35] put forward a method based on cosine similarity to measure the similarity between texts. Improved square-cosine similarity

is applied to various document understanding tasks, such as text classification, clustering and query search. And compared with existing methods, the experimental results showed that the method was more effective. Similarity calculation is also used in image processing [36]. An image registration method based on mutual information and homograph theory is proposed in Ref. [37]. This method used mutual information as the similarity measurement criterion, and homograph theory as the integrity transformation model, to improve the solution of homograph matrix. The experimental results suggested that the algorithm has good accuracy and robustness.

It can be seen that similarity calculation has broad prospects [38,39], but few researchers have attempted to connect similarity calculation with redundancy analysis between features, which is a topic that should be discussed and explored further.

3. Preliminaries

In this section, we illustrate some basic preliminaries about feature selection using mutual information strategy. The symbols are uniformly defined as shown in Table 1. Where, the symbol F represents the feature set, Y represents the category label, and the symbol S represents the selected feature subset.

3.1. FS based on MI strategy

Information theory was first used to find the value relationship between data compression and transmission rate under the background of communication theory. With the emergence of FS algorithm, the probability measurement criterion in information theory is applied to the FS measurement standard, which is mostly used to quantify the uncertainty between features and labels, and has achieved very good results. MI is a measure of the amount of information a random variable has about another variable, which is a basic concept in information theory. It provides a method to quantify the correlation of feature subsets with respect to the output vector Y , which is very useful in FS.

In the process of feature selection, let x be a characteristic variable after the discretization of feature set X , n be the number of categories after the discretization of data set X . The MI of X and Y is defined as follows:

$$I(x; y) = \sum_{i=1}^n \sum_{j=1}^n p(x(i), y(j)) \log_2 \left(\frac{p(x(i), y(j))}{p(x(i))p(y(j))} \right) \quad (1)$$

where $p(x(i)) = P\{x = x(i), x(i) \in X\}$, represents the probability density of X . Entropy can be interpreted as the negative expectation of the logarithm of probability distribution.

Selecting mechanism is the core of FS method. A good filtering mechanism can usually take into account the number and performance of feature subsets. The model proposed in this paper uses

MI strategy to filter the correlation between features and labels. The implementation process is shown in Algorithm 1.

The basic idea is to use forward search to calculate the MI $I(X_i, Y)$ between feature X_i and label Y (refer to formula 1), and use this as a measurement criterion to find the most similar feature between the candidate feature and the sample category.

Algorithm 1 Mutual information sorting algorithm

Input: Original data set F ; Category label Y
Output: Sorted data set S

- 1: $D = \text{Discrete}(F)$; //Discrete dataset F
- 2: $N = \text{Size}(D)$; //Calculate the number of features
- 3: $n = \text{Size}(Y)$; //Calculate the number of labels
- 4: **while** $X \in D \& i \leq n$
- 5: $\text{totsamples} \leftarrow \text{histc}(x_i)$; //Statistical feature distribution frequency
- 6: $P_{x(i)} \leftarrow \frac{\text{cumsum}(\text{totsamples})}{n}$; //Calculate the probability of feature distribution within an interval
- 7: $y \leftarrow \frac{\text{cumsum}(Y)}{n}$; //Calculate the distribution probability of different label categories
- 8: $I(X; Y) \leftarrow \sum_{i=1}^n \sum_{j=1}^n p(x(i), y(j)) \log_2 \left(\frac{p(x(i), y(j))}{p(x(i))p(y(j))} \right)$; //Calculate MI of features
- 9: $I[N] \leftarrow I(X; Y)$;
- 10: **end while**
- 11: $\text{idx}[N] \leftarrow \text{sort}(I[N])$; //Sort the MI for each feature

The classical FS algorithms generally select the features with the top k MI values of features and labels as the key feature subset to complete the FS process. This kind of algorithm has the advantages of simplicity, high efficiency, low time complexity and small amount of computation. However, the algorithm only considers the correlation between candidate features and category labels, and ignores the redundancy between features. It is easy to over fit when constructing classification model for training. Therefore, many algorithms mine the correlation between features and labels, and at the same time, remove redundancy. Typical algorithms include MIM, mRMR, CMIM, CIFE, etc. These algorithms calculate the MI between features to determine whether they are redundant. Due to the influence of data discretization in the preprocessing stage, there will be a large deviation in judging the degree of redundancy between different features, which will ultimately affect the effect of FS.

3.2. Two-dimensional Cosine transform

Cosine transform belongs to orthogonal transform, which uses cosine function to express data information. Cosine transform is a kind of block transform, which is mostly used for image data, such as image compression, speech signal processing and so on. From the point of view of signal processing, cosine transform can transform the signal from time domain to frequency domain, and has good decorrelation performance.

The cosine transform is used to analyze the feature data for the first time. The high-frequency information of the feature data is obtained through the cosine transform, which is used to judge the similarity between the features and remove the redundant features. In real life, the most intuitive way to compare the similarities and differences between different features is to convert feature information into "photos", and eliminate redundant features by analyzing the similarity between "photos". For this purpose, we use two-dimensional cosine transform to extract feature information. Let $x(i, j)$ be the value of the coordinates (i, j) in the feature

Table 1
Notation description.

Symbol	Description
F	Original feature set
Y	Category label
S	Selected feature set
$\text{Discrete}(F)$	Discretization F
X_i	A candidate feature $X_i \in F$
X_s	A selected feature $X_s \in S$
$P_{x(i)}$	The frequency of occurrence of i in feature x
$I(X; Y)$	The MI between two variables
$SU(X; Y)$	Calculation of symmetric uncertainty for X and Y
$\text{idx}(N)$	Position coordinate after sine sorting of N
$\text{Sim}(X_i; X_j)$	Calculate the similarity between two variables

variable x , and x be a square matrix of size $N * N$, then the two-dimensional cosine transform of $x(i, j)$ is defined as:

$$CT(u) = \begin{cases} \sqrt{\frac{1}{N}} \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} x(i, j) \cos\left[\frac{(i+0.5)\pi}{N} u\right] \cos\left[\frac{(j+0.5)\pi}{N} v\right] & \text{if } u = 0 \\ \sqrt{\frac{2}{N}} \sum_{i=0}^{N-1} \sum_{j=0}^{N-1} x(i, j) \cos\left[\frac{(i+0.5)\pi}{N} u\right] \cos\left[\frac{(j+0.5)\pi}{N} v\right] & \text{if } u \neq 0 \end{cases} \quad (2)$$

where, $CT(u, v)$ is the coefficient after cosine transform. The information in the original data after cosine transform is centrally displayed in the upper left part of the matrix after cosine transform. One dimensional cosine transform can only extract the relevant information of a single sequence. Compared with a single dimension, the information reflected by two-dimensional data is more intuitive, which is also more suitable for people's intuitive feeling. Therefore, we prefer to use two-dimensional cosine change analysis data to obtain more information between features, when judging the similarity of features.

4. Proposed method for feature selection

A FS algorithm based on feature matrix similarity relation and MI, it's structure is shown in Fig. 1. Firstly, the features are discretized, and the MI between data features and tags is calculated in turn. Secondly, the single feature variable is transformed into a two-dimensional matrix, and the feature is cosine transformed to extract the key information of the feature. Thirdly, according to the key information of different features, the acquaintance relationship between different feature matrices is calculated, and the similarity is calculated according to the distance between the key information. Fourthly, the MI values between features and tags are sorted, and the order of key features is obtained by dynamically adjusting the similarity between features. Finally, the optimal feature subset is obtained according to the order of key features, and the FS process is completed.

4.1. Definition of feature matrix similarity

To solve the problem of key information loss due to discretization, we propose the matrix operation of feature variables to judge the similarity of different matrices, which is used as a criterion to measure the redundancy between different features. Fig. 1 shows the overall process of similarity calculation in the dotted box.

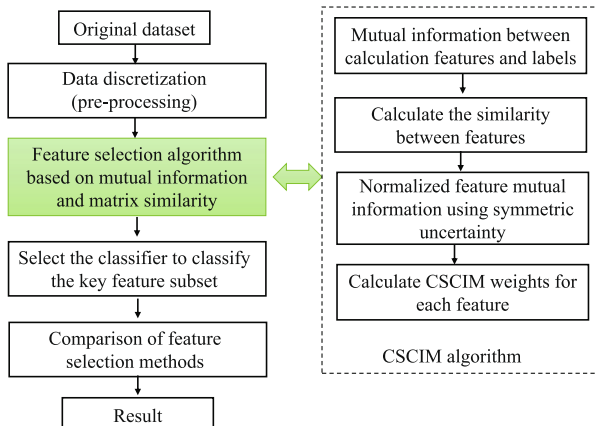


Fig. 1. Basic structure block diagram.

4.1.1. Two-dimensional matrix transformation of feature sequences

Different from the comparison between single feature variables, the two-dimensional matrix can better display the differences between features, and the similarities and differences between features can be better displayed. Therefore, we have constructed a feature transformation method, as shown in Definition 1.

Definition 1. Two-dimensional matrix transformation of feature sequences. Given a feature X , let $x(i)$ be the value of the i th in the feature variable X , N be the length of the feature X , that is, $x = \{x(1), x(2), \dots, x(N)\}$, the two-dimensional transformation matrix is TX , and the feature matrix can be converted into the following formula:

$$TX = \begin{bmatrix} x(1) & x(\lceil \sqrt{N} \rceil + 1) & \dots & x(\lceil \sqrt{N} \rceil - 1 * \lceil \sqrt{N} \rceil + 1) \\ x(2) & x(\lceil \sqrt{N} \rceil + 2) & \dots & x(\lceil \sqrt{N} \rceil - 1 * \lceil \sqrt{N} \rceil + 2) \\ \dots & \dots & \dots & \dots \\ x(\lceil \sqrt{N} \rceil) & x(2 * \lceil \sqrt{N} \rceil) & \dots & x(N) \end{bmatrix} \quad (3)$$

In particular, when $\sqrt{N}$ is not an integer, the TX matrix is filled with 0.

Example 4.1. Let the original data feature have M feature variables, and each variable is composed of N samples. Obtain each feature of the original data in turn, and convert it into a square matrix with equal rows and columns. If the data is insufficient, fill it with 0. The process of converting the original feature into a two-dimensional matrix is shown in Fig. 2.

4.1.2. Feature similarity calculation

After transforming a single data feature into a two-dimensional matrix, it is necessary to compare the differences between features in turn and analyze the acquaintance relationship between features. We first apply the two-dimensional cosine transform to the feature matrix to obtain the information of the feature high-frequency components. Secondly, the two-dimensional discrete cosine transform features are sampled, the high-frequency region in the upper left corner of the matrix is extracted, and the mean value in the sampling region is calculated. Thirdly, set the part larger than the mean value in the sampling area to 1 and the rest to 0 to construct a hash fingerprint. Finally, the hash fingerprints of different feature vectors are calculated in turn, and the distance between different hash fingerprints is calculated. The ratio of distance and sampling size is taken as the similarity between features.

According to formula 3, transform the original feature sequence to obtain a feature matrix with equal rows and columns, and then perform cosine transform on the feature matrix (refer to formula 2). Extract the hash fingerprint matrix from the transformed matrix, and its detailed definition is as follows.

Definition 2. Hash fingerprint matrix. It is known that SP is a high-frequency information matrix intercepted by a cosine transformed feature square matrix, and the hash fingerprint matrix is defined as follows:

$$hash = \begin{cases} SP = 1, SP > means(SP) \\ SP = 0, SP \leq means(SP) \end{cases} \quad (4)$$

where $means$ is the mean value in the matrix region.

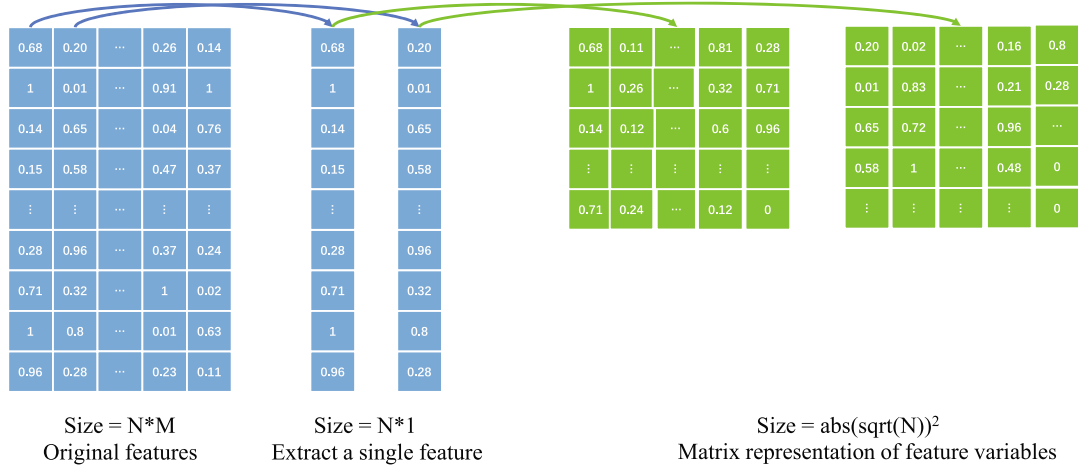


Fig. 2. Process diagram of converting original features to two-dimensional matrix.

Definition 3. Similarity calculation. It is known that *hash1* and *hash2* are hash fingerprint matrices after cosine transform. The similarity of the two matrices is expressed as $\text{Similarity}(\text{hash1}; \text{hash2})$. The definition formula is as follows:

$$\text{Sim}(\text{hash1}; \text{hash2}) = \frac{(\text{sample} - \text{distance}(\text{XOR}(\text{hash1}, \text{hash2})))}{\text{sample}} \quad (5)$$

where *distance* is the Hamming distance after the exclusive or of *hash1* and *hash2*. *Sample* is the sampling size, which is generally taken as half of the initial matrix size.

Example 4.2. Let the two feature matrices be *Matrix1* and *Matrix2* respectively. The size of the two matrices is 4×4 . The process of similarity calculation is shown in Fig. 3. The sample size in the figure is 2×2 , that is, *sample* = 2. The ratio of sampling scale minus Hamming distance and sampling size is used to measure the similarity between the two vectors. The similarity between *Matrix1* and *Matrix2* in the figure is 0.5.

4.1.3. FS algorithm based on matrix similarity and MI

The traditional FS method relies on MI whether it is to judge the correlation between variables and labels or the redundancy between variables. As explained above, after discretizing the char-

acteristic variables, a lot of information will be lost, which will affect the effect of the algorithm. Therefore, we bring in cosine similarity to overcome the information loss caused by MI.

Definition 4. Evaluation function of key feature of CSCIM_FS algorithm. Let X_i and X_s respectively represent a candidate feature and a FS in the original feature set, S is the selected feature subset, and Y is the label corresponding to the feature. Then the evaluation function formula of CSCIM_FS algorithm is defined as follows:

$$J_{\text{CSCIM}}(X_i) = SU(X_i; Y) - \frac{1}{|S|} \sum_{X_s \in S} \text{Sim}_{\text{normal}}(\text{hash}(X_s; \text{hash}(X_i))) \quad (6)$$

where, $SU(X_i; Y)$ represents the symmetric uncertainty relationship between X_i and Y . Due to the difference of MI between different features and labels after discretization, we use SU transform to normalize the MI between features and labels. The formula is as follows:

$$SU = \frac{2 * I(X_i; Y)}{H(X_i) + H(Y)} \quad (7)$$

At the same time, to solve the problem that the MI strategy does not match the similarity measurement standards between features, we normalize the similarity between features to $0 \sim 1$. The calculation formula is as follows:

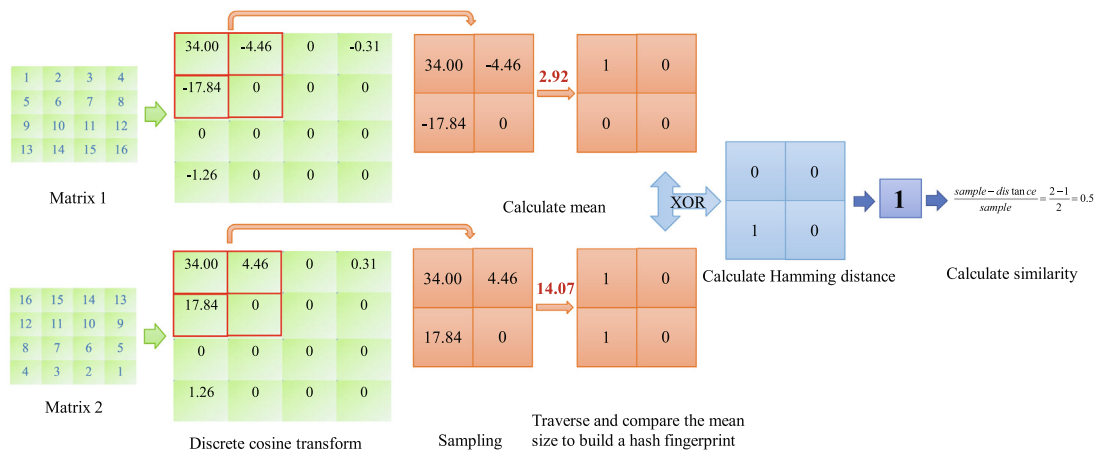


Fig. 3. Similarity calculation process.

$$Sim_{normal} = 2 * \frac{Sim - Sim_{min}}{Sim_{max} - Sim_{min}} - 1 \quad (8)$$

The implementation process of CSCIM_FS algorithm is shown in Algorithm 2.

Algorithm 2 CSCIM_FS algorithm

Input: Original data set F ; Category label Y ;
 MI set between features and labels $I[N]$; Number of key feature subsets k
Output: Feature subset S

```

1:  $S = \emptyset$ ; // Initialization feature subset
2:  $SU[N] = SU(I)$ ; // Calculate symmetry uncertainty between features and labels
3:  $i = 1; j = 1$ ;
4: while  $X_i \in F \& X_j \in F \& i \leq N \& j \leq N$  do
5:    $Sim[N] \leftarrow Sim(X_i, X_j)$ ; // Calculate the similarity between features
6:    $i++; j++$ ;
7: end while
8:  $Sim = Normalized(Sim)$ ; // Normalize  $Sim$  to between 0 ~ 1
9:  $idx = \max(SU)$ ; // Find the maximum value in  $SU$ 
10:  $i = 1$ ;
11: while  $i \leq N$  do
12:   if  $SU[i] - Sim \geq idx$  then
13:      $idx \leftarrow SU[i]$ ;
14:   end if
15:    $i++$ ;
16: end while
17:  $j = 1$ ;
18: while  $k \leq N$  do
19:    $S[j] \leftarrow idx[j]$ ;
20:    $j++$ ;
21: end while
```

The features of data can be roughly divided into three main types. 1. relevant features: it is helpful for learning tasks and can improve the effect of classification algorithm; 2. irrelevant features: it does not help the learning task and will not bring any improvement to the effect of the algorithm; 3. redundant feature: it will not bring new information to the task, or the information of this feature can be inferred from other features. The main function of the FS algorithm is to build an evaluation function to find out the features most related to the category label, and eliminate the features that are irrelevant and redundant with the selected variables. Among them, finding the redundancy between features is one of the most difficult processes for FS algorithm. We use the similarity between two-dimensional matrices as the measurement criterion to improve the drawback of over reliance on information measurement criterion in the process of removing redundancy. The theory proposed in this section will be verified by experiments.

5. Experiments and analysis

5.1. Experimental framework

The algorithm verification experiment is carried out on Windows system matlab2020a. Several common FS algorithms such as: MIM, SD (statistical dependency), QPFS (rectangular programming), mRMR [26], maxrel (maximum correlation) [19], minred (minimum redundancy), MIQ (mutual information quotation), CMIM, CIFE, MCRMC, DCSF (Dynamic Change of Selected Feature with the class) [40], UCRFS (Feature Selection considering Uncer-

tainty Change Ratio) [41], etc. are compared in turn. Several classifiers are used to classify feature subsets. The experimental process is shown in Fig. 4.

5.1.1. Data sets

To verify the effectiveness of CSCIM_FS, the experiments are implemented on 12 benchmark data sets that are from the UCI repository. From the UCI database, multiple groups of data with more than 10 dimensions are selected, covering image, biomedicine, climate and other fields, including binary classification and multi classification data sets. The description of these data sets is listed in Table 2.

5.1.2. Multiple classifiers

Typically, researchers put results into classifiers that have validated the performance of FS methods. In this paper, we use four classifiers, i.e. K-nearest neighbors(KNN), decision tree(DT), random forest(RF) and support vector machines(SVM), to verify our method [42,43].

- KNN is a supervised learning method. Find the optimal classification accuracy by adjusting the number of the most similar data K in the KNN algorithm.
- DT algorithm is also a common classifier, which is suitable for solving classification problems. The algorithm can accurately sort the categories. The working principle of the algorithm is similar to the flow chart. The data points are divided into two similar categories at one time, from “trunk” to “branch” and then to “leaf”, so that these categories become more similar in a limited range, and then complete the data classification.
- RF is one of the probabilistic supervised learning methods used for classification. The advantage is that there are few adjustment parameters, only the number of different trees T needs to be set. It can show excellent classification performance on multiple data sets.
- SVM is a discriminative supervised learning method. The optimal classification accuracy can be obtained by changing different kernel functions and adjusting the penalty factor C .

Classifiers are mostly used to verify the advantages and disadvantages of FS algorithms. The simpler the classifier is, the better. By default, KNN classifiers are often used. At the same time, it is also necessary to consider the differences between different features and the correlation between feature attributes. Therefore, we will compare multiple classifiers to jointly verify the quality of the FS algorithm.

5.1.3. Performance metrics

After the data is selected and classified, it is necessary to build appropriate evaluation indicators to evaluate the effectiveness of the method [44]. Although the commonly used accuracy index is an important and non negligible index, it may be misleading when evaluating the uneven distribution of samples or small sample data. Therefore, we use multiple indexes together to judge the effect of the algorithm.

To fully verify the accuracy of the model, k -fold cross validation is used to cross validate the classified data for small sample data sets. K -fold cross validation is the most common cross validation method. The basic idea is to divide the data set into k parts, form complementary sets for the validation set and the test set in turn, keep cycling, and get the average results. The general formula for cross validation is $\bar{R} = \frac{1}{k} \sum_{i=1}^k R_i$, where R_i is the result of the i -th cross validation of each round, and $\bar{R}$ is the average result of k -fold cross validation.

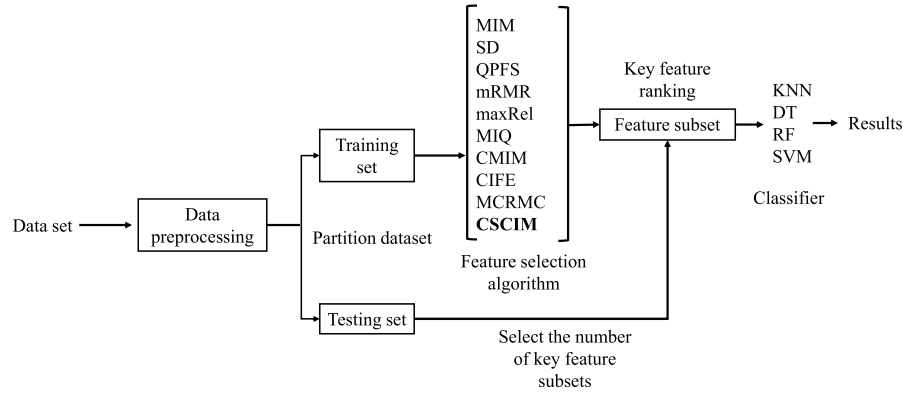


Fig. 4. Experimental flow chart.

Table 2
Description of data sets.

ID	Dataset Name	Size	Features	Class
1	Sonar	208	60	2
2	Lonosphere	351	34	2
3	WDBC	569	30	2
4	Madelon	2600	500	2
5	Air	369	64	3
6	DNAtest	1186	180	3
7	Lung	203	3312	5
8	GSE72526	61	800	4
9	SRBCT	83	2308	4
10	Brain	90	5920	5
11	ORL	400	1024	40
12	Isolet	1560	617	26

	Predict		
	Confusion matrix	P	N
	P	True Positives(TP)	False Negatives(FN)
	N	False Positives(FP)	True Negatives (TN)

Fig. 5. Confusion matrix diagram.

As mentioned earlier, the most common way to deal with classification problems is the confusion matrix. As shown in Fig. 5, the confusion matrix can represent most of the evaluation indicators. According to the characteristics of different data types, this paper uses four statistical evaluation indexes to evaluate the performance of the algorithm. They are the accuracy, precision, recall and F1 values of classification [45]. The description of each index is shown in Table 3.

5.2. Experimental analysis

To verify the effectiveness and interpretability of the algorithm, we conducted experiments on Fisher data sets by synthesizing Gaussian mixture data. The initial data set has 4 features, 3 categories and 150 samples in total. We use the covariance matrix to add Gaussian noise to expand the data dimension. By means of random replacement, a data set consisting of 100 related and 100 unrelated features is constructed. Category labels are represented by red, green and blue respectively, and display all features of the data. As shown in Fig. 6.

Table 3
Evaluation index description.

Name	Formula Definition
Accuracy	$Accuracy = \frac{TP+TN}{TP+FP+TN+FN} * 100\%$
Precision	$Precision = \frac{TP}{TP+FP} * 100\%$
Recall	$Recall = \frac{TP}{TP+FN} * 100\%$
F1_score	$F1_score = 2 * \frac{Precision * Recall}{Precision + Recall} * 100\%$

FS is carried out for the data in turn, 5-fold cross validation is adopted, and KNN classifier is used to classify the feature subsets. The number of key feature subsets ranges from 1 to 30. The experimental results are shown in the Fig. 7.

We have compared various baselines, and the red broken line in Fig. 7 means the best results. To verify the classification effect of CSCIM_FS algorithm, we also use the confusion matrix of classification results as shown in Fig. 8.

From the comparison chart of confusion matrix, we can see that our algorithm has obvious advantages over other algorithms. Calculate the precision, recall and F1_score of each category in turn, and the results are shown in Fig. 9.

In the above experiments, the number of key feature subsets was all 10. We compared 10 feature subsets obtained under different FS algorithms, and used 4 classifiers for classification. From Fig. 10, we find that the classification accuracy is different under different classifiers and the FS algorithm still has the greatest impact on the classification accuracy. Compared with other algorithms, CSCIM_FS algorithm can well complete the selecting of the key features and have stronger competitiveness. In the next experiment, we will test the algorithm on the UCI public data set to verify its practicability in the real environment.

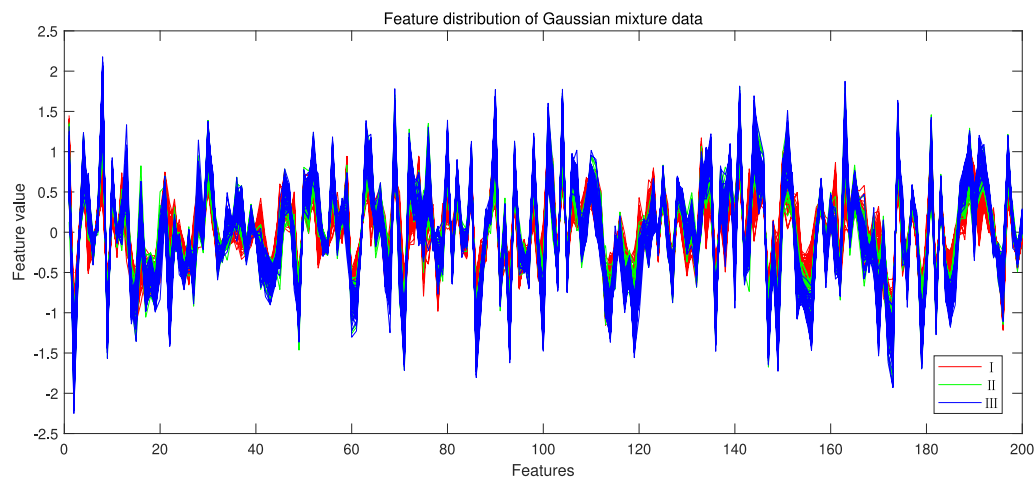


Fig. 6. Features distribution diagram.

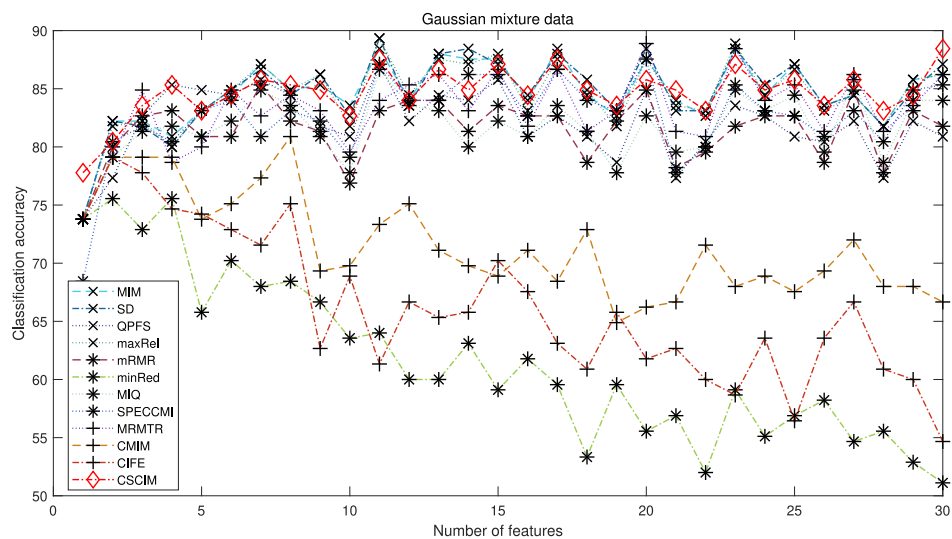


Fig. 7. Experimental result.

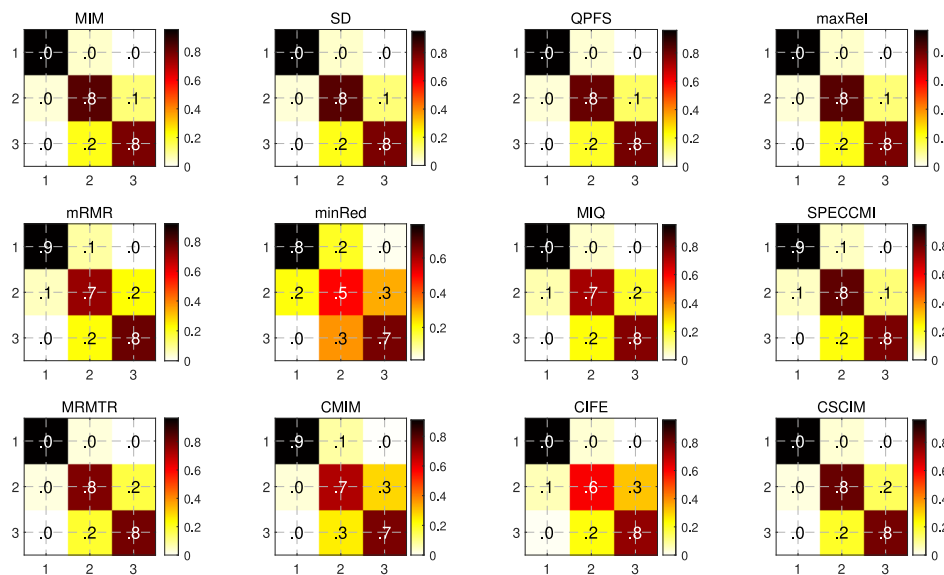


Fig. 8. Confusion matrix comparison experiment results.

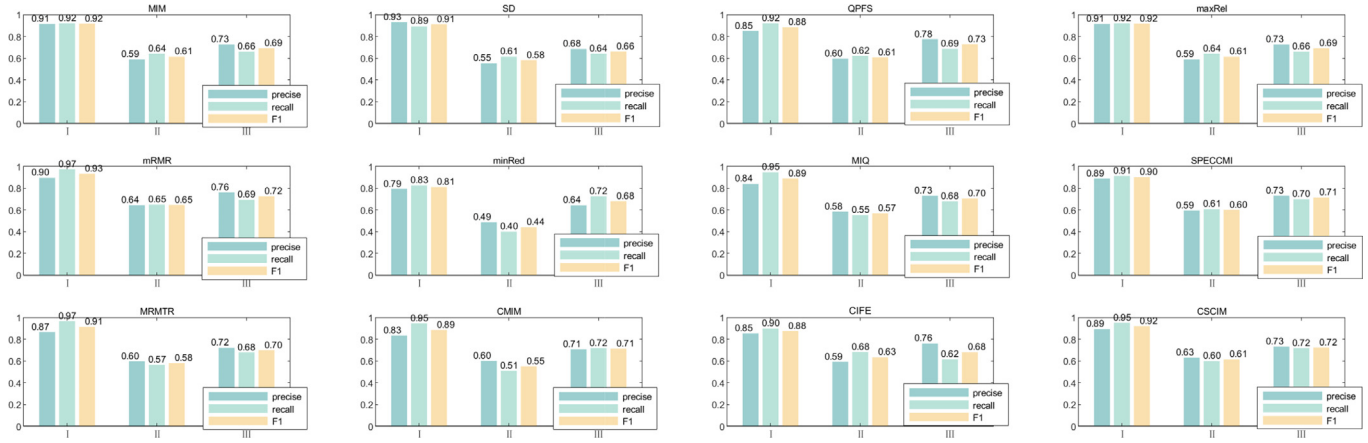


Fig. 9. Comparison chart of evaluation indicators for experimental results of various algorithms.

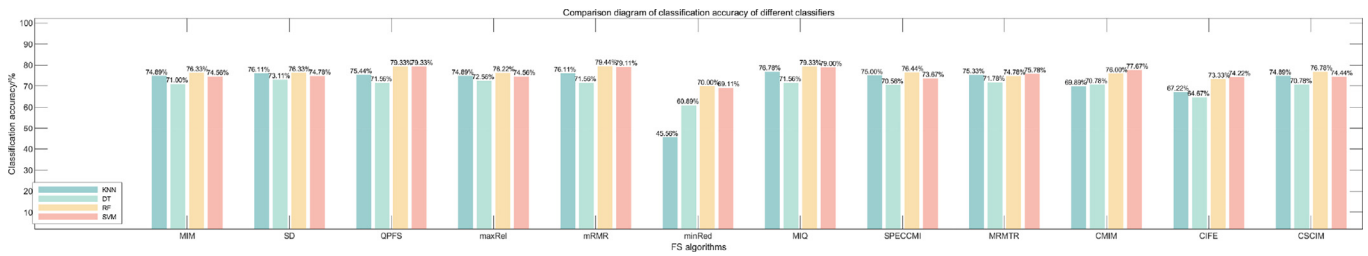


Fig. 10. Comparison of classification accuracy of multiple FS algorithms in different classifiers.

5.3. Comparison results

We select several UCI public data sets, compared multiple FS methods, and conducted experiments with 5-fold cross validation. Selecting a different number of key feature subsets, and the experimental results are shown in Fig. 11.

The four experiments in the figure above are all conducted on two-category data. Among them, the data dimensions of Fig. 11a, Fig. 11b and Fig. 11c are below 100 dimensions, and the data dimensions of Fig. 11d are as high as 500 dimensions. It shows that when the dimension is less than 100, the CSCIM_FS algorithm has obvious advantages, and when the dimension is more than 100, the effect is slightly lacking. Futuremore, we conducted tests on multiple multi-categorical data sets and the results are shown in Fig. 12.

The datasets in Fig. 12a and Fig. 12b are air datasets and DNA datasets respectively, which correspond to three categories, with feature dimensions of 64 and 180. Fig. 12c and Fig. 12d are two medical data, with feature dimensions exceeding 500. Fig. 12e and Fig. 12f are two multi classification datasets with dimensions above 1000. With the increase of the number of key features, the classification accuracy of CSCIM_FS algorithm is gradually improved. It can be seen that CSCIM_FS algorithm not only has good results, but also has very good stability. We use the above 10 data sets, with 20 key feature subsets, and use 5-fold cross validation to compare the classification accuracy of 10 FS algorithms. The experimental results are shown in Table 4. Compared with the experimental results in the table, it can be concluded that although the accuracy of CSCIM_FS algorithm is not as good as other methods on some data, the feature subset obtained by this algorithm is more stable [46]. Compared with other algorithms, the feature subset obtained by CSCIM_FS algorithm has obvious advantages.

In order to verify the classification accuracy of the feature subsets obtained by the above 10 FS algorithms under other classifiers,

we selected three classifiers DT, RF and SVM to classify the feature subsets. The experimental results are shown in Table 5–7.

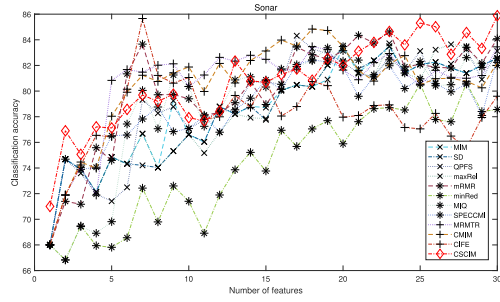
Compared with the experimental results in the above three tables, we find that the feature subset obtained by CSCIM_FS algorithm has obvious advantages when the classifiers are the same.

We perform feature selection on unbalanced data to test the advantages and disadvantages of CSCIM_FS algorithm in processing unbalanced data. We take the Lung data set as an example for verification. The data set has 3312 features, 203 samples and 5 categories. Table 8 describes the distribution of the five categories of the data set. According to the description, the Lung data set is a typical unbalanced data set. When building a classification model for data, we should focus on II, IV and V categories, and we select 30 feature subsets and use KNN classifier to classify the feature subsets. Compared with multiple FS algorithms, the confusion matrix of the experimental results is shown in Fig. 13.

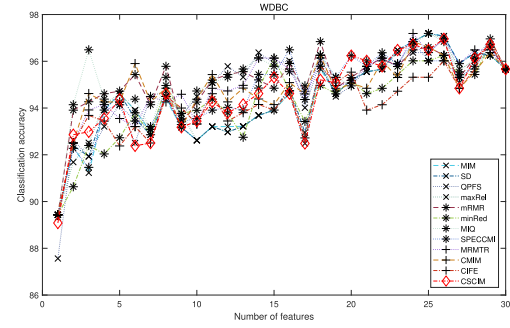
A total of five FS algorithms are compared in the above experiments. It can be seen that CSCIM_FS algorithm still has great advantages in the face of unbalanced data. Calculating the precision, recall and F1_score of the five categories of lung data in turn, and the results are shown in Fig. 14. Comparing the three evaluation indexes of Lung dataset classification, it is not difficult to find that CSCIM_FS algorithm is also suitable for feature selection of unbalanced data.

After completing the testing on UCL public data, we tested the effectiveness of the CSCIM_FS algorithm on multi category data (ORL) and Isolet. In addition to comparing common FS methods, we also compared some new FS methods, such as DCSF, UCRFS, etc [40,41]. The results are shown in Table 9.

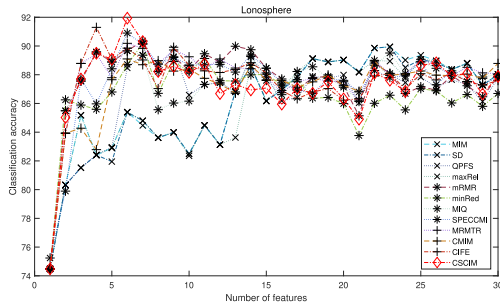
Otherwise, to verify the effectiveness of the CSCIM_FS algorithm in removing redundant features, we select the Short-duration heavy rainfall (SDHR) dataset from a practical application scenario. The short-term heavy precipitation observation data of 13 meteo-



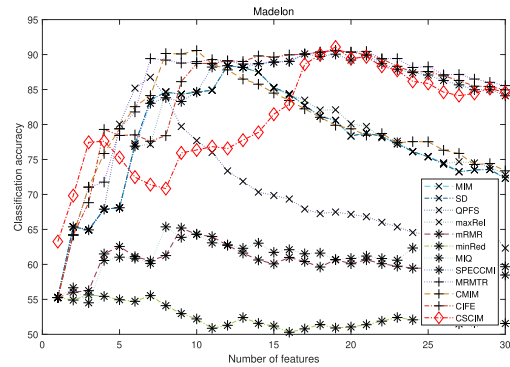
(a) Sonar dataset



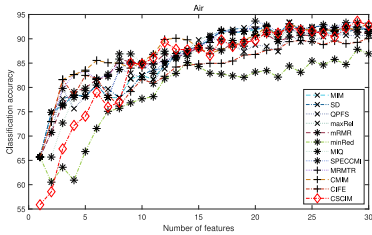
(b) WDBC dataset



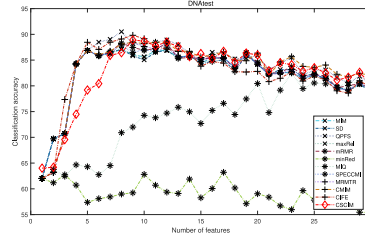
(c) Lonosphere dataset



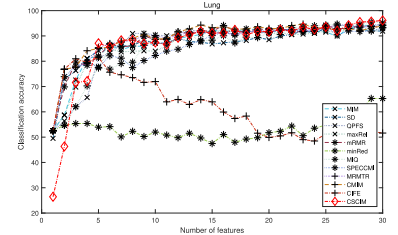
(d) Madelon dataset

Fig. 11. Average classification accuracy of the KNN classifier for each of the 4 two-category datasets.

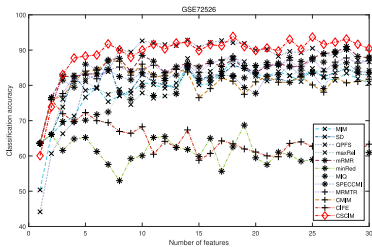
(a) Air dataset



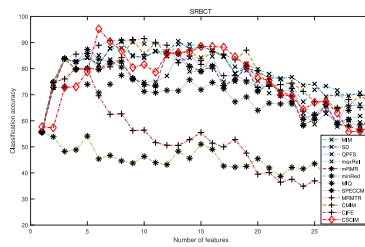
(b) DNAtest dataset



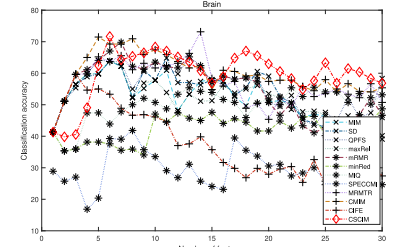
(c) Lung dataset



(d) GSE72526 dataset



(e) SRBCT dataset



(f) Brain dataset

Fig. 12. Average classification accuracy of the KNN classifier for each of the 4 multi-categorical datasets.

rological stations in East China were collected, including data with 54 features of 19 main factors, including convective suppression energy, convective effective energy, and upper air water vapor flux. The dataset contains 3,838 records, of which the ratio of normal events to SDHR events is 3,253:585.

We use KNN classifier and 30 features in SDHR dataset to test CSCIM_FS algorithm. The experimental results are shown in Fig. 15. It can be seen that the accuracy of the features obtained by CSCIM_FS algorithm is better than other FS algorithms in classification.

Table 4
The KNN predictive accuracy of CSCIM_FS VS. competing algorithms.

Dataset	MIM	SD	QPFS	mRMR	maxRel	MIQ	CMIM	CIFE	MCRMC	CSCIM_FS
Sonar	83.46%	83.46%	84.51%	84.02%	83.95%	83.94%	82.97%	82.64%	84.73%	84.85%
Lonosphere	89.58%	89.92%	87.73%	87.25%	89.58%	88.54%	88.82%	88.63%	89.49%	89.77%
WDBC	95.13%	95.13%	95.46%	95.43%	95.13%	95.46%	95.66%	95.34%	95.31%	95.34%
Madelon	78.34%	78.34%	66.82%	61.60%	78.92%	61.08%	75.85%	90.23%	90.23%	82.77%
Air	90.15%	91.73%	88.20%	88.99%	92.33%	88.99%	91.26%	85.88%	92.33%	93.66%
DNAtest	84.95%	84.90%	85.49%	84.94%	84.95%	75.69%	86.25%	83.04%	84.95%	86.09%
Lung	91.70%	91.22%	89.16%	88.33%	88.33%	90.39%	85.39%	81.44%	87.17%	91.93%
GSE72526	67.95%	61.31%	66.31%	79.88%	74.06%	72.67%	70.11%	34.26%	79.26%	79.31%
SRBCT	71.10%	65.96%	80.71%	71.44%	75.06%	71.44%	76.10%	38.41%	76.94%	77.69%
Brain	49.62%	51.11%	54.62%	54.25%	46.85%	59.44%	40.92%	28.70%	53.14%	59.48%
AVG	80.19%	79.30%	79.90%	79.61%	80.91%	78.76%	79.33%	70.85%	83.35%	84.08%

Table 5
The DT predictive accuracy of CSCIM_FS VS. competing algorithms.

Dataset	MIM	SD	QPFS	mRMR	maxRel	MIQ	CMIM	CIFE	MCRMC	CSCIM_FS
Sonar	75.42%	76.17%	76.41%	77.36%	75.19%	75.03%	77.75%	80.00%	83.79%	84.31%
Lonosphere	91.60%	90.60%	90.41%	92.36%	92.17%	90.98%	90.93%	91.22%	91.79%	92.46%
WDBC	94.66%	94.58%	94.63%	94.75%	95.25%	94.52%	94.46%	94.46%	94.69%	95.52%
Madelon	79.21%	78.92%	70.01%	62.48%	78.94%	62.94%	76.85%	85.18%	85.05%	86.58%
Air	85.57%	84.68%	84.73%	84.82%	85.66%	85.10%	86.45%	86.17%	85.51%	86.71%
DNAtest	90.47%	90.64%	90.80%	91.03%	90.68%	85.21%	90.97%	91.03%	90.55%	91.99%
Lung	87.27%	88.59%	86.79%	88.36%	87.03%	88.27%	87.44%	85.81%	87.36%	87.93%
GSE72526	62.26%	64.13%	66.11%	68.02%	70.16%	64.86%	65.48%	41.33%	67.98%	68.07%
SRBCT	71.04%	73.09%	71.09%	73.79%	71.00%	72.60%	72.47%	60.41%	70.06%	72.64%
Brain	75.74%	76.66%	74.62%	74.25%	75.37%	74.07%	76.66%	69.44%	76.29%	76.11%
AVG	81.32%	81.80%	80.80%	80.72%	82.14%	79.35%	81.94%	78.50%	83.30%	84.23%

Table 6
The RF predictive accuracy of CSCIM_FS VS. competing algorithms.

Dataset	MIM	SD	QPFS	mRMR	maxRel	MIQ	CMIM	CIFE	MCRMC	CSCIM_FS
Sonar	73.60%	73.60%	72.88%	73.20%	73.20%	72.88%	76.17%	74.47%	73.68%	75.84%
Lonosphere	83.74%	84.50%	84.35%	86.11%	83.74%	85.83%	84.74%	83.93%	83.59%	87.69%
WDBC	94.72%	94.72%	95.42%	95.51%	94.72%	95.10%	95.42%	95.45%	95.28%	95.39%
Madelon	59.99%	59.99%	60.23%	60.62%	59.71%	60.54%	59.97%	59.53%	59.53%	60.96%
Air	80.30%	82.25%	80.66%	80.11%	81.13%	80.11%	80.84%	75.41%	81.13%	81.32%
DNAtest	91.64%	91.52%	92.62%	91.57%	91.64%	86.57%	92.66%	91.33%	91.64%	92.95%
Lung	93.27%	94.91%	95.416%	94.17%	94.17%	95.24%	96.72%	91.96%	94.42%	96.70%
GSE72526	78.62%	72.83%	79.45%	79.19%	75.59%	73.97%	80.59%	35.42%	78.93%	79.03%
SRBCT	95.09%	95.43%	95.59%	94.62%	96.19%	94.62%	96.34%	70.61%	98.20%	98.81%
Brain	84.81%	83.33%	81.11%	81.48%	81.48%	84.07%	79.25%	62.77%	80.55%	84.29%
AVG	83.57%	83.30%	83.77%	83.65%	83.15%	82.89%	84.27%	74.08%	83.69%	85.29%

Table 7
The SVM predictive accuracy of CSCIM_FS VS. competing algorithms.

Dataset	MIM	SD	QPFS	mRMR	maxRel	MIQ	CMIM	CIFE	MCRMC	CSCIM_FS
Sonar	77.50%	77.50%	75.99%	75.98%	77.50%	75.66%	78.15%	74.80%	76.31%	78.23%
Lonosphere	84.56%	85.33%	86.04%	85.61%	84.56%	84.52%	84.61%	84.71%	85.42%	86.33%
WDBC	96.69%	96.69%	96.86%	96.89%	96.69%	96.86%	96.48%	96.04%	97.24%	97.51%
Madelon	61.13%	61.12%	61.11%	60.89%	61.10%	60.89%	60.85%	61.74%	61.74%	62.42%
Air	71.06%	70.41%	73.11%	74.03%	71.06%	74.03%	74.31%	56.75%	71.06%	74.94%
DNAtest	92.78%	93.26%	93.46%	92.98%	92.78%	89.36%	93.52%	93.28%	92.78%	93.53%
Lung	91.49%	91.33%	91.83%	91.25%	91.25%	91.26%	92.73%	88.58%	92.08%	92.78%
GSE72526	80.32%	79.76%	82.73%	84.10%	81.14%	77.79%	80.86%	41.47%	84.94%	84.68%
SRBCT	94.99%	94.17%	95.39%	93.17%	94.98%	93.17%	93.20%	70.24%	97.79%	96.23%
Brain	72.96%	73.14%	72.40%	72.22%	74.44%	72.77%	71.66%	67.59%	73.33%	73.00%
AVG	82.34%	82.27%	82.89%	82.71%	82.55%	81.63%	82.63%	73.52%	83.26%	83.96%

To verify the advantages of CSCIM_FS algorithm in removing redundant features, we compared the performance of MI algorithm and CSCIM_FS algorithm on SDHR dataset. Select the top 15 key features for comparison. From Fig. 16, it can be seen that in

Fig. 16a, the redundant features obtained using the MI strategy include QFLUX, THETA, DPT, Q300, and Q500. Combined with weight variance, the difference in similar features is relatively small. And our method, with only DIV and RH indices for redun-

Table 8

Lung dataset description.

Category	Number of samples	Sample proportion
I	139	68.47%
II	17	8.37%
III	21	10.34%
IV	20	9.85%
V	6	2.96%

dant features, combined with the classification accuracy in Fig. 15, can conclude that the CSCIM algorithm has significant advantages.

Through the experimental comparison, it can be seen that CSCIM_FS algorithm can achieve very good results in both two-category data and multi-categorical data. The feature similarity calculated by cosine transform combined with mutual information strategy can improve the traditional FS method which only relies on MI. The experiment proved that our scheme is feasible.

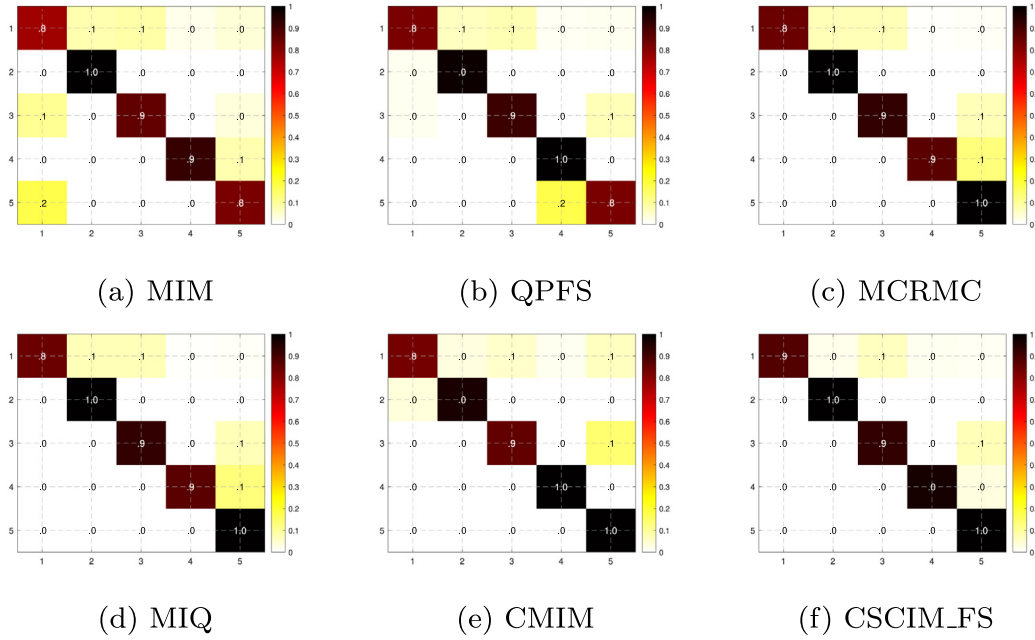


Fig. 13. Lung dataset classification confusion matrix results.

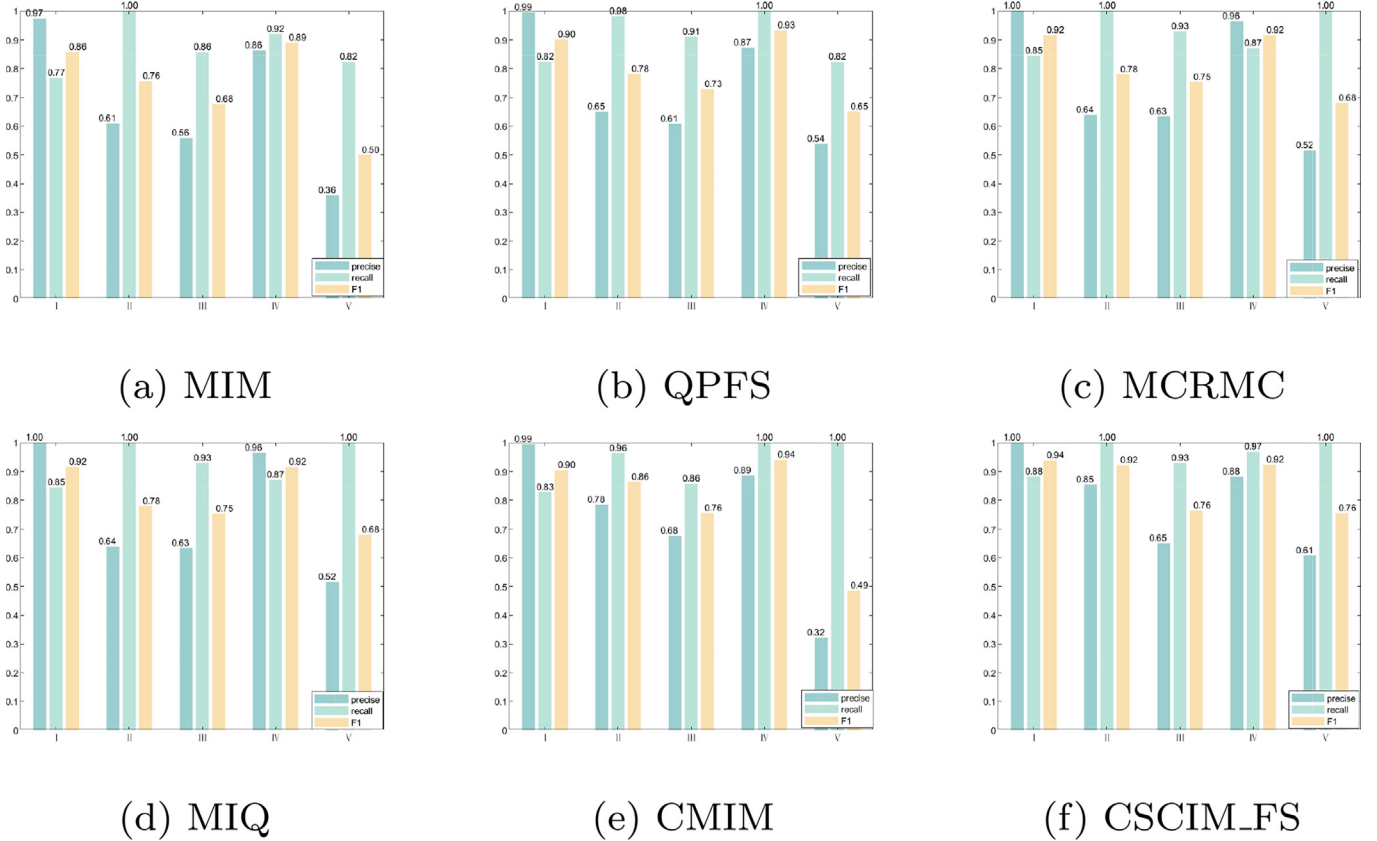


Fig. 14. Evaluation of classification results of Lung dataset.

Table 9

The predictive accuracy of CSCIM_FS vs. competing algorithms (%).

Datasets	Methods	Accuracy
ORL	JMI	78.25
	mRMR	75.75
	CIFE	64.50
	DCSF	69.13
	UCRFS	82.50
	CSCIM_FS	83.12
Isolet	JMI	65.19
	mRMR	65.13
	CIFE	70.10
	DCSF	74.36
	UCRFS	75.67
	CSCIM_FS	76.77

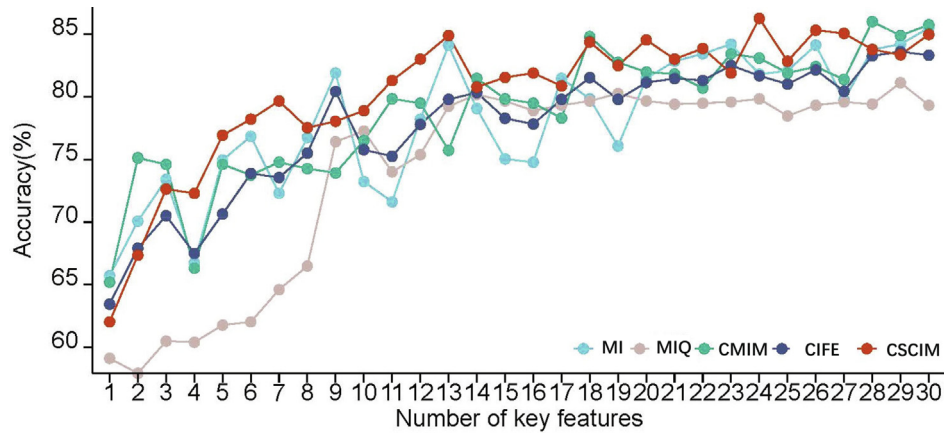
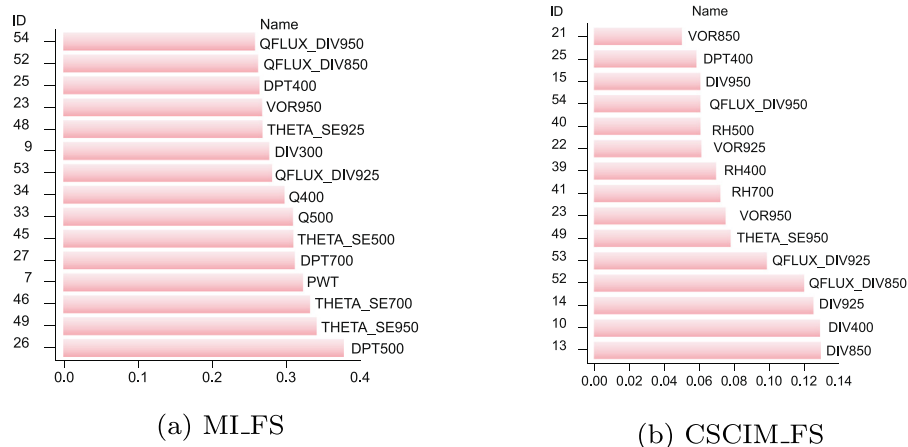
5.4. Discussion

Compared with multiple FS algorithms, CSCIM_FS algorithm has obvious advantages in average accuracy and stability. The data sets used in the above experiments are all continuous data sets. Traditional feature selection algorithms based on mutual information theory (such as MIM, CMIM, mRMR, etc.) need to discretize the data, which is bound to lose a lot of information. If the FS algorithm based on information theory is not used, it is difficult to find the relationship between different features and labels, which affects the accuracy of FS methods to a certain extent. CSCIM_FS algorithm

can calculate the similarity based on the high-frequency components obtained by cosine transform, which can just solve the problem of information lost in feature discretization. Moreover, the algorithm calculates the similarity between features by constructing a two-dimensional feature matrix, which avoids the repeated calculation of multiple features and improves the efficiency of the algorithm. Therefore, it has obvious advantages in FS of continuous data.

6. Conclusion and future research

CSCIM_FS based on the cosine similarity coefficient and information measurement criterion is proposed. Compared with other methods, this method has the advantages of high accuracy and high stability in feature selection of continuous data sets. The traditional feature selection based on mutual information strategy needs to discretize the data when preprocessing the data to facilitate the calculation of the relationship between features. This process will cause a lot of information loss. However, few people pay attention to this problem and carry out targeted research on it. To solve this problem, this paper proposes a similarity strategy between features to replace the traditional mutual information measurement criteria. This scheme can solve the problem of information loss when the feature is discretized. Moreover, this paper proposes a two-dimensional feature matrix strategy, which transforms the original one-dimensional feature sequence into a two-

**Fig. 15.** SDHR dataset experimental result.**Fig. 16.** Comparison of the top 15 key feature subsets.

dimensional matrix. This strategy can compare the relationship between features from multiple angles and directions, and has very obvious advantages in feature selection of data.

To evaluate the classification performance of CSCIM_FS, we have carried out experiments on several data, and the results show that the scheme proposed in this paper is feasible. Furthermore, we analyze the effect of this method in dealing with unbalanced data. The results show that CSCIM_FS algorithm can also perform well in dealing with unbalanced data.

Future work will include multivariable correlation feature selection. At present, CSCIM_FS algorithm only considers two features when measuring the similarity between features. However, in reality, many data categories are often determined by multiple characteristics, such as a person's age, gender and weight, which can determine the person's health status. Therefore, it is necessary to consider the correlation between multivariable features when selecting features from data [47,48]. Moreover, CSCIM_FS algorithm can be combined with multiple application scenarios, such as judging tumor markers and identifying key factors in meteorological data. Continuously improve the accuracy of data classification.

CRedit authorship contribution statement

Gaoteng Yuan: Conceptualization, Methodology, Software, Formal analysis, Writing - original draft. **Yi Zhai:** Writing - review & editing, Funding acquisition. **Jiansong Tang:** Writing - review & editing. **Xiaofeng Zhou:** Supervision.

Data availability

Data will be made available on request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported partly by Graduate research and innovation projects in Jiangsu Province (KYCX22-0609), the National Natural Science Foundation of China under Grant (11901325).

References

- [1] Y. Wang, X. Li, R. Ruiz, Feature selection with maximal relevance and minimal supervised redundancy, *IEEE Trans. Cybern.* 7 (1) (2022) 1–11, <https://doi.org/10.1109/TCYB.2021.3139898>.
- [2] S. Lall, D. Sinha, A. Ghosh, D. Sengupta, S. Bandyopadhyay, Stable feature selection using copula based mutual information, *Pattern Recogn.* 112 (2021), <https://doi.org/10.1016/j.patcog.2020.107697>.
- [3] G. Sun, J. Li, J. Dai, Z. Song, F. Lang, Feature selection for iot based on maximal information coefficient, *Future Generat. Comput. Syst.* 89 (DEC) (2018) 606–616, <https://doi.org/10.1016/j.future.2018.05.060>.
- [4] A. Jayaprakash, C. Keziselvavijila, Feature selection using ant colony optimization (aco) and road sign detection and recognition (rsdr) system, *Cognit. Syst. Res.* 58 (DEC) (2019) 123–133, <https://doi.org/10.1016/j.cogsys.2019.04.002>.
- [5] S. Lall, D. Sinha, A. Ghosh, D. Sengupta, S. Bandyopadhyay, Stable feature selection using copula based mutual information, *Pattern Recogn.* 112 (1) (2020), <https://doi.org/10.1016/j.patcog.2020.107697>.
- [6] Y. Li, T. Li, H. Liu, Recent advances in feature selection and its applications, *Knowl. Inf. Syst.* 53 (3) (2017) 551–577, <https://doi.org/10.1007/s10115-017-1059-8>.
- [7] J. Maldonado, M.C. Riff, B. Neveu, A review of recent approaches on wrapper feature selection for intrusion detection, *Expert Syst. Appl.* 198 (2022), <https://doi.org/10.1016/j.eswa.2022.116822>.
- [8] S. Kashef, H. Nezamabadi-pour, A label-specific multi-label feature selection algorithm based on the pareto dominance concept, *Pattern Recogn.* 88 (2019) 654–667, <https://doi.org/10.1016/j.patcog.2018.12.020>.
- [9] Y. Lin, Q. Hu, J. Liu, J. Duan, Multi-label feature selection based on max-dependency and min-redundancy, *Neurocomputing* 168 (2015) 92–103, <https://doi.org/10.1016/j.neucom.2015.06.010>.
- [10] H. Yan, Q.L. Ye, D.J. Yu, Efficient and robust twsvm classification via a minimum l1-norm distance metric criterion, *Mach. Learn.* 108 (6) (2019) 993–1018, <https://doi.org/10.1007/s10994-018-5771-8>.
- [11] G. Li, Y. Li, Y. Zheng, Y. Li, X. Zhou, A novel feature selection approach with pareto optimality for multi-label data, *Appl. Intell.* 51 (8) (2021) 7794–7811, <https://doi.org/10.1007/s10489-021.02228-2>.
- [12] L. Ling, L. Huawen, M. Zongjie, M. Yuchang, D. Zhengjie, Z. Jiaqing, Z. Jianmin, Multi-label feature selection via information gain, *Adv. Data Mining Appl.* 8933 (2014) 345–355, <https://doi.org/10.1007/978-3-319-14717-8.27>.
- [13] J. González, J. Ortega, M. Damas, P. Martín-Smith, J.Q. Gan, A new multi-objective wrapper method for feature selection - accuracy and stability analysis for bci, *Neurocomputing* 333 (2019) 407–418, <https://doi.org/10.1016/j.neucom.2019.01.017>.
- [14] Y. Ying, W. Yinglong, Feature selection for multi-label learning using mutual information and ga, in: *Rough Sets and Knowledge Technology*, Springer International Publishing, Cham, 2014, pp. 454–463. doi:10.1007/978-3-319-11740-9.42.
- [15] J. Zhang, Z. Luo, C. Li, C. Zhou, S. Li, Manifold regularized discriminative feature selection for multi-label learning, *Pattern Recogn.* 95 (2019) 136–150, <https://doi.org/10.1016/j.patcog.2019.06.003>.
- [16] J. Yin, T. Tao, J. Xu, A multi-label feature selection algorithm based on multi-objective optimization, in: *2015 International Joint Conference on Neural Networks IJCNN 2015 Killarney Ireland July 2015*, 2015, pp. 1–7. doi:10.1109/IJCNN.2015.7280373.
- [17] Z. Peng, W.B. Ni, Z. Shu, Online group streaming feature selection considering feature interaction, *Knowl.-Based Syst.* 226 (10715) (2021) 1–11, <https://doi.org/10.1016/j.knosys.2021.107157>.
- [18] X. Li, J. Ren, Micq-ippo: An effective two-stage hybrid feature selection algorithm for high-dimensional data, *Neurocomputing* 501 (2022) 328–342, <https://doi.org/10.1016/j.neucom.2022.05.048>.
- [19] Y. Lin, Q. Hu, J. Liu, J. Duan, Multi-label feature selection based on max-dependency and min-redundancy, *Neurocomputing* 168 (2015) 92–103, <https://doi.org/10.1016/j.neucom.2015.06.010>.
- [20] Y. Qu, R. Li, A. Deng, C. Shang, Q. Shen, Non-unique decision differential entropy-based feature selection - sciencedirect, *Neurocomputing* 393 (14) (2020) 187–193, <https://doi.org/10.1016/j.neucom.2018.10.112>.
- [21] S. Sharmin, M. Shoyaib, A.A. Ali, M. Khan, O. Chae, Simultaneous feature selection and discretization based on mutual information, *Pattern Recogn.* 91 (16) (2019) 162–174, <https://doi.org/10.1016/j.patcog.2019.02.016>.
- [22] Y. Zhang, Y. Ma, Non-negative multi-label feature selection with dynamic graph constraints, *Knowl.-Based Syst.* 238 (2022), <https://doi.org/10.1016/j.knosys.2021.107924>.
- [23] P. Zhang, W. Gao, J. Hu, Y. Li, A conditional-weight joint relevance metric for feature relevancy term, *Eng. Appl. Artif. Intell.* 106 (2021), <https://doi.org/10.1016/j.engappai.2021.104481>.
- [24] D.J. Nicolas, P.-C. Simon, O. Catharina, E.-H. Nehme, B. Gianluca, H.-K. Benjamin, mrmre: an r package for parallelized mrmr ensemble feature selection, *Bioinformatics* 29 (18) (2013) 2365–2368, <https://doi.org/10.1093/bioinformatics/btt383>.
- [25] B. R. Using mutual information for selecting features in supervised neural net learning, *IEEE Trans. Neural Networks* 5(4) (1994) 537–550. doi:10.1109/72.298224.
- [26] H. Peng, F. Long, C. Ding, Feature selection based on mutual information: Criteria of max-dependency, max-relevance, and min-redundancy, *IEEE Trans. Pattern Anal. Mach. Intell.* 27 (8) (2005) 1226–1238, <https://doi.org/10.1109/TPAMI.2005.159>.
- [27] H. Yang, J. Moody, Data visualization and feature selection: New algorithms for nongaussian data, *Adv. Neural Inform. Process. Syst.* 12.
- [28] D. Lin, X. Tang, Conditional infomax learning: An integrated framework for feature extraction and fusion, in: *Computer Vision ECCV 2006*, Springer, Berlin Heidelberg, Berlin, Heidelberg, 2006, pp. 68–82, <https://doi.org/10.1007/11744023.6>.
- [29] M. Bennasar, Y. Hicks, R. Setchi, Feature selection using joint mutual information maximisation, *Expert Syst. Appl.* 42 (22) (2015) 8520–8532, <https://doi.org/10.1016/j.eswa.2015.07.007>.
- [30] H. Li, W. Wang, Z. Liu, Y. Niu, H. Wang, S. Zhao, Y. Liao, W. Yang, X. Liu, A novel locality-sensitive hashing relational graph matching network for semantic textual similarity measurement, *Expert Syst. Appl.* (2022), <https://doi.org/10.1016/j.eswa.2022.117832>.
- [31] W. Hu, L. Wu, M. Jian, Y. Chen, H. Yu, Cosine metric supervised deep hashing with balanced similarity, *Neurocomputing* 448 (2021) 94–105, <https://doi.org/10.1016/j.neucom.2021.03.093>.
- [32] W. Gao, L. Hu, Y. Li, P. Zhang, Preserving similarity and staring decisis for feature selection, *IEEE Trans. Artif. Intell.* 2 (6) (2021) 584–593, <https://doi.org/10.1109/TAI.2021.3105084>.
- [33] Y. Kai, G. Gong-De, L. Jing, S. Lin, Quantum algorithms for similarity measurement based on euclidean distance, *Int. J. Theor. Phys.* 59 (10) (2020) 3134–3144, <https://doi.org/10.1007/s10773.020.04567.1>.

- [34] R. Lin, B. Wu, Y. Su, An adaptive weighted pearson similarity measurement method for load curve clustering, *Energies* 11 (9) (2018) 2466, <https://doi.org/10.3390/en11092466>.
- [35] S. Sahar, W. Dingding, Improved sqrt-cosine similarity measurement, *J. Big Data* 4 (25) (2017) 2196, <https://doi.org/10.1186/s40537-017-0083-6>.
- [36] D. Li, J. Li, Y. Fan, G. Lu, J. Ge, X. Liu, Printed label defect detection using twice gradient matching based on improved cosine similarity measure, *Expert Syst. Appl.* 204 (2022), <https://doi.org/10.1016/j.eswa.2022.117372>.
- [37] Y.E. Cheng, Y.J. Qian, Image registration based on mutual information and corner points, *Product Design and Manufacturing*, vol. 338, of *Advanced Materials Research*, 2011, pp. 267–271, <https://doi.org/10.4028/www.scientific.net/AMR.338.267>.
- [38] D.F. Chipier, A structured fast algorithm for the vlsi pipeline implementation of inverse discrete cosine transform, *Circuits Syst. Signal Process.* 40 (9) (2021) 36, <https://doi.org/10.1007/s00034-021-01718-5>.
- [39] A.E. Hassani, B.A. Skourt, A. Majda, Efficient lung nodule classification method using convolutional neural network and discrete cosine transform, *Int. J. Adv. Comput. Sci. Appl.* 12 (2) (2021) 766–768, <https://doi.org/10.14569/IJACSA.2021.0120296>.
- [40] W. Gao, L. Hu, P. Zhang, Class-specific mutual information variation for feature selection, *Pattern Recogn.* 79 (2018) 328–339, <https://doi.org/10.1016/j.patcog.2018.02.020>.
- [41] P. Zhang, W. Gao, Feature selection considering uncertainty change ratio of the class label, *Appl. Soft Comput.* 95 (2020), <https://doi.org/10.1016/j.asoc.2020.106537>.
- [42] H.C. Vieira, J. Santos, D.V. Souza, P. Rios, S. Nisgoski, Near-infrared spectroscopy for the distinction of wood and charcoal from fabaceae species: comparison of ann, knn and svm models, *Forest Syst.* 29 (3) (2020), <https://doi.org/10.5424/fs/2020293-16965>.
- [43] A. Mc, J. Yuan, A. Cg, A. Hl, Y.A. Wei, Learning features from enhanced function call graphs for android malware detection, *Neurocomputing* 423 (7) (2021) 301–307, <https://doi.org/10.1016/j.neucom.2020.10.054>.
- [44] W. Gao, L. Hu, P. Zhang, F. Wang, Feature selection by integrating two groups of feature evaluation criteria, *Expert Syst. Appl.* 110 (2018) 11–19, <https://doi.org/10.1016/j.eswa.2018.05.029>.
- [45] Z.C. Lipton, C. Elkan, B. Naryanaswamy, Optimal thresholding of classifiers to maximize f1 measure, in: *Machine Learning and Knowledge Discovery in Databases*, Springer, Berlin Heidelberg, Berlin, Heidelberg, 2014, pp. 225–239, https://doi.org/10.1007/978-3-662-44851-9_15.
- [46] P. Zhou, N. Wang, S. Zhao, Online group streaming feature selection considering feature interaction, *Knowl.-Based Syst.* 226 (2021), <https://doi.org/10.1016/j.knosys.2021.107157>.
- [47] P. Qiu, Z. Niu, Tcic_fs: Total correlation information coefficient-based feature selection method for high-dimensional data, *Knowl.-Based Syst.* 231 (2021), <https://doi.org/10.1016/j.knosys.2021.107418>.
- [48] K. Yu, X. Guo, L. Liu, J. Li, H. Wang, Z. Ling, X. Wu, Causality-based feature selection: Methods and evaluations, *ACM Comput. Surv.* 53 (5) (2020) 36, <https://doi.org/10.1145/3409382>.



Yi Zhai received his Ph.D. degree in Louisiana State University Center for Health Sciences, Biostatistics in 2017. Currently, she is a lecturer at the School of Computer Science and Technology, Qilu University of Technology (Shandong Academy of Sciences), China. Her research areas are artificial intelligence and biometrics.



Jiansong Tang received his MS degree in the School of Electronic Science and Engineering, Nanjing University in 2020. He is working toward the Ph.D. degree in the college of computer and information, Hohai University. His research interests include image processing and pattern recognition.



Xiaofeng Zhou He is currently a professor at the college of computer and information in Hohai University. His current interests are in the areas of image processing, intelligent information processing, pattern recognition, etc.



Gaoteng Yuan received his MS degree in the School of Computer Science and Technology from Qilu University of Technology (Shandong Academy of Sciences) in 2020. He is working toward the Ph.D. degree in the college of computer and information, Hohai University. His research interests include image processing, feature selection and information theory.